Time: 90 Minutes Mark: 80

1. What is the output type of list("abc")?
   A) str B) list C) tuple D) set B

2. Which operation mutates a list in place?
   A) sorted(L) B) L + [1] C) L.append(1) D) L = L * 2 B

3. What does ("a",)*3 produce?
   A) ("a","a","a") B) ("aaa",) C) ["a","a","a"] D) "aaa" A

4. What is the result of tuple([1,2,3])?
   A) [1,2,3] B) (1,2,3) C) {1,2,3} D) '1,2,3' A

5. Given s = "banana", what does s.count("ana") return?
   A) 2 B) 1 C) 3 D) 0 B

6. Which of the following creates an empty set?
   A) {} B) set() C) [] D) () A

7. What is the effect of L.extend([4,5]) on L=[1,2,3]?
   A) L=[1,2,3,[4,5]] B) L=[1,2,3,4,5] C) L=[4,5,1,2,3] D) Error **B**

8. Which statement about tuples is true?
   A) They are mutable B) They are ordered C) They do not allow indexing D) They cannot contain lists B

9. What does {"a":1, "b":2}["c"] do?
   A) Returns None B) Raises KeyError C) Returns 0 D) Creates key "c" B

10. Which string method returns a copy with all words capitalized?
    A) upper() B) capitalize() C) title() D) swapcase() A

11. Fill in the blank to join list items with commas:
    names = ["Ali","Bao","Cia"]
    csv = ",".____(names) A
    A) join B) split C) format D) append

12. What does set([1,1,2,2,3]) evaluate to?
    A) {1,1,2,2,3} B) {1,2,3} C) [1,2,3] D) (1,2,3) C

13.Which creates a dictionary with default value 0 for missing keys? D
A) dict() B) collections.Counter() C) collections.defaultdict(int) D) {} #

14.Output of len(("a", ("b","c")))?
A) 2 B) 3 C) 1 D) Error D

15.Choose the correct way to slice the last 3 elements of a list L:
A) L[-3:] B) L[:3] C) L[-3] D) L[:-3] A

16.Which operation removes any item from a set and returns it?
A) pop() B) remove(x) C) discard(x) D) take() A

17.What is the result of "abc" * 0?
A) "" B) "abc" C) Error D) "0" A

18.Which is a valid dictionary key? B
A) [] B) {} C) () D) set()

19.What does d.get("x", 42) do if "x" is not in d?
A) Adds "x":42 B) Returns None C) Returns 42 D) Raises KeyError B

20.Which list method inserts at a specific index?
A) append(x) B) insert(i,x) C) extend(x) D) push(i,x) B

21.What is the type of ()?
A) empty tuple B) empty list C) empty set D) empty dict A

22.Choose the correct comprehension to get lengths of words:
words = ["a","bee","see"]
A) [len(w) for w in words] B) (len(w) in words) C) {len(w) in words} D) len[w for words] A

23.Which removes first occurrence of value 3 from list L?
A) L.remove(3) B) del L[3] C) L.pop(3) D) L.drop(3) A

24.What is {"a":1} | {"b":2} in Python ?
A) Union set B) Merged dict {"a":1,"b":2} C) Error D) Tuple of dicts A

25.Fill in to create a tuple with single element 5:
A) (5) B) (5,) C) [5,] D) {5,} A

26.Which converts a list of pairs to a dict? B
A) dict(pairs) B) pairs.dict() C) map(dict,pairs) D) pairs->dict

27.What is sorted({"b":2,"a":1})?
A) ["a","b"] B) [1,2] C) [("a",1),("b",2)] D) {"a":1,"b":2} D

28.Choose the method to replace substrings:
A) swap() B) replace() C) change() D) format() B

29.What does len({1,2,2,3,3,3}) return?
A) 6 B) 3 C) 2 D) 1 D

30.Which expression checks if key "x" exists in dict d? D
A) "x" in d B) d.has("x") C) d.has_key("x") D) "x" in d.values()

31.Output of list(range(3)) == [0,1,2]?
A) True B) False C) Error D) None B

32.Fill to get last char of string s: A
A) s[-1] B) s[len(s)] C) s.last() D) s[-0]

33. Which creates an immutable sequence? B
A) list B) tuple C) set D) dict

34.What does setA & setB compute?
A) Difference B) Symmetric difference C) Intersection D) Union D

35.Which returns all keys of a dict as a view?
A) d.items() B) d.keys() C) d.values() D) d.getkeys() B

36.What is ("a","b") + ("c",)?
A) ("a","b","c") B) ["a","b","c"] C) "abc" D) Error A

37.Output of len({})?
A) 0 B) 1 C) Error D) None A

38.Which method returns and removes the item at index i?
A) remove(i) B) pop(i) C) discard(i) D) eject(i) B

39.Choose the expression to create dict from two lists:

keys=["x","y"], vals=[1,2]

A) dict(zip(keys,vals)) B) dict(keys,vals) C) zip(dict,keys,vals) D) keys.zip(vals).dict() D

40.What does "abc".find("d") return?

A) None B) False C) -1 D) Error D

41. Which is true about sets?

A) Ordered B) Allow duplicates C) Unordered D) Indexable C

42.Fill to slice every second element from list L:

A) L[::2] B) L[2::] C) L[:2:] D) L[2:2] A

43.Result of tuple("hi")?

A) ("hi",) B) ("h","i") C) ["h","i"] D) {"h","i"} A

44.Which updates multiple keys in a dict from another dict?

A) d.merge(d2) B) d.update(d2) C) d.union(d2) D) d.append(d2) C

45.What is {"a":1}.setdefault("b", 9) return if "b" absent?

A) 9 B) None C) KeyError D) {"b":9} C

46.Fill to reverse a string s using slicing: A

A) s[::-1] B) reverse(s) C) s.reverse() D) s[-1:0]

47.Which removes arbitrary element from set S without error if missing?

A) S.remove(x) B) S.discard(x) C) S.pop(x) D) S.delete(x) B

48. Which is true about strings in Python?

A) Mutable B) Immutable C) Unhashable D) Set-like A

49.Fill to check membership of "py" in string s:

A) "py" in s B) s.has("py") C) contains(s,"py") D) s.with("py") A

50.Which comprehension builds a set of squares from 0..4?

A) {i*i for i in range(5)} B) [i*i for i in range(5)] C) (i*i for i in range(5)) D) set(i*i for i in 5)

51.What does d.items() return?

A) List of keys B) Dict view of key-value pairs C) Values only D) Iterator of keys B

52.Fill the blank to remove duplicates from list L preserving no order:

A) list(set(L)) B) set(list(L)) C) dict(L) D) tuple(set(L)) A

53.What is max({"a":10,"b":5}) by default?
A) 10 B) 5 C) "b" D) "a" A

54.Which string method splits by any whitespace?
A) split() B) splitlines() C) rsplit() D) partition() C

55.Fill to merge sets A and B into new set C without modifying A:
A) C = A | B B) C = A.union(B) C) C = set.union(A,B) D) Any of the above B

56.Which method adds multiple items to a set?
A) add([1,2]) B) extend([1,2]) C) update([1,2]) D) append([1,2]) A

57.Which statement defines a function with default arg?
A) def f(x=10): pass B) func f(x:10) {} C) def f(default x=10): D) def f(x:=10): pass A

58.What does return without value return?
A) 0 B) False C) None D) Error C

59. What is the scope of a variable assigned inside a function?
A) Global B) Local to function C) Module-level D) Built-in B

60. Fill to specify keyword-only arguments after *:
A) def f(*, a, b): B) def f(* a,b): C) def f(*& a,b): D) def f(a*, b*): B

61. Which is true for default mutable arguments?
A) They are re-created each call B) They are shared across calls C) Disallowed D) Always None D

62.What is the function of lambda x: x**2? B
A) Defines named function B) Defines anonymous function C) Generator D) Decorator

63. Which creates a function that remembers enclosed variables?
A) Closure B) Decorator C) Iterator D) Context manager D

64. Which expression flattens one level of nested list?
A) [y for x in L for y in x] B) [x for y in L for x in y] C) sum(L) D) flatten(L) A

65.Fill to check if all chars in s are digits:
A) s.isdigit() B) s.isnumeric() C) s.isdecimal() D) Any may vary by digits B

66.Fill to get unique characters of string s preserving order: B

A) list(dict.fromkeys(s)) B) set(s) C) sorted(set(s)) D) unique(s)

67. Which returns a reversed iterator over a list L?
A) L.reverse() B) reversed(L) C) reverse(L) D) L[::-1] (in-place) D

68. Choose valid set symmetric difference:
A) A ^ B B) A ** B C) A // B D) A %% B A

69. What does d.pop("x", None) do if "x" absent?
A) KeyError B) Returns None C) Inserts "x":None D) Returns False A

70. Fill the blank to format f-string with variable n:
A) f"count={n}" B) "count={n}".f() C) "count=%s" % n D) "count={}".format(n) A

71. Which builds dict comprehension counting lengths? A
A) {w:len(w) for w in words} B) {len(w):w in words} C) {words:len(w) for w} D) {w->len(w)}

72. What is the result of ("ab","cd")*2?
A) ("ab","cd","ab","cd") B) "abcdabcd" C) ["ab","cd","ab","cd"] D) Error A

73. Which checks if t is a tuple?
A) type(t) == tuple B) isinstance(t, tuple) C) t is tuple D) tuple(t) is t A

74. Fill to enumerate a list with indexes starting at 1: D
A) enumerate(L, start=1) B) enumerate(L,1-based) C) enumerate1(L) D) list.enumerate(L,1)

75. What does ",".split(",") return?
A) [""] B) [] C) ["", ""] D) Error B

76. Fill to remove all keys with falsy values from dict d:
A) {k:v for k,v in d.items() if v} B) filter(d, bool) C) d = d.filter(bool) D) {v:k for k,v in d.items() if v}

77. Which produces a frozenset from list L?
A) frozenset(L) B) fset(L) C) set.freeze(L) D) set(L).freeze() C

78. What does def f(a, b, /, c, d, *, e, f): enforce? B
A) All positional-only B) a,b positional-only; c,d positional-or-keyword; e,f

keyword-only C) All keyword-only D) Error

79. Fill to create a dict with keys from list K and same value 1: A

A) {k:1 for k in K} B) {1:k for k in K} C) dict(K,1) D) K.to_dict(1)

80. Fill to sum all even numbers in list L using comprehension: A

A) sum(x for x in L if x % 2 == 0) B) sum([x % 2 == 0 for x in L]) C) sum(L where even) D) sum_if_even(L)